

PAPER NAME

Smartgtid_Paper_Final.pdf

AUTHOR

-

WORD COUNT

6504 Words

CHARACTER COUNT

36842 Characters

PAGE COUNT

8 Pages

FILE SIZE

1.8MB

SUBMISSION DATE

Aug 14, 2026 3:31 PM GMT+1

REPORT DATE

Aug 14, 2026 3:32 PM GMT+1

● 24% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

- 24% Internet database
- 0% Publications database
- 0% Submitted Works database

Smart Grid Sensor Network for Industrial Energy Management Using IoT and Machine Learning

Saeed Sarwar Anas

MSc Computer Science and Technology

Ulster University, UK

Anas-SS@ulster.ac.uk

Abstract—Industrial facilities account for a substantial share of global electricity demand, yet most small and medium-sized enterprises (SMEs) operate without any real-time visibility of their energy use because commercial monitoring infrastructure is prohibitively expensive. This paper presents the design and evaluation of a low-cost smart grid sensor network for industrial energy management that integrates Internet of Things (IoT) sensing with machine learning. The system couples an inexpensive embedded sensing platform with a reproducible data pipeline, a Long Short-Term Memory (LSTM) model for short-term demand forecasting, and an Isolation Forest model for unsupervised anomaly detection, all surfaced through a real-time dashboard. Using the UCI Individual Household Electric Power Consumption dataset, the LSTM forecaster achieved a mean absolute error (MAE) of 0.3707 kW, outperforming a naive baseline by 8.6% and a 24-hour moving-average baseline by 35.7% on unseen test data. Cross-dataset validation on the Tetouan City load confirmed that the unchanged pipeline transfers across scales, improving on the same baselines by 62.0% and 87.0%. The evaluation uses two public benchmark datasets, with deployment of the designed sensing layer reserved for future work. The contribution is an affordable, open-source, end-to-end pipeline that lowers the barrier to intelligent energy management for resource-constrained organisations.

Index Terms—smart grid, Internet of Things, machine learning, LSTM, energy demand forecasting, anomaly detection, isolation forest, edge computing, Industry 4.0

I. INTRODUCTION

The electricity grid is among the most important engineering systems of the modern era, but for most of the twentieth century it operated as an essentially “one-way” network. Electricity flowed from a small number of large, central generators to passive consumers, and the grid carried almost no information back about how, when, or where energy was being used. This architecture was robust and well suited to the demand patterns of its time, yet it left operators fundamentally blind to consumption at the point of use and unable to respond to changing conditions in real time.

Several converging pressures over the past two decades have exposed the limits of this traditional model. Global electricity demand has continued to rise, with industry responsible for a large fraction of total consumption [1]. At the same time, decarbonisation targets and the rapid uptake of intermittent renewable generation have demanded a far more flexible and observable grid. These pressures gave rise to the concept of the *smart grid*: an electricity network augmented with digital sensing, two-way communication, and data-driven control,

so that supply and demand can be monitored and balanced dynamically rather than managed reactively. Surveys of smart grid communication infrastructure document this shift from a passive network towards a sensor-rich, communication-enabled system [2], [3].

Three paradigms have shaped the sector since. The first is the move from centralised generation towards *distributed* energy resources, which multiplies the points that must be observed and coordinated [4]. The second is *Industry 4.0* and the Industrial Internet of Things, in which low-cost networked sensors make pervasive, fine-grained measurement economically feasible [5]. The third is *machine learning*, which has moved energy systems from threshold-based monitoring towards predictive management: deep learning models nonlinear patterns in sequential data [6], recurrent architectures such as the Long Short-Term Memory (LSTM) network [7] match the daily and weekly cycles of consumption, and unsupervised methods such as the Isolation Forest [8] flag abnormal usage without the labelled fault data that is rarely available in practice. The combined impact is significant: sensor-driven feedback has been shown to reduce consumption [9], [10], LSTM forecasters to outperform classical statistical baselines [11], [12], and anomaly detection to surface equipment faults before they escalate [13].

Despite this progress, a clear gap remains. Each of the four integrated energy-management platforms reviewed in Section IV [4], [14]–[16] depends on proprietary or edge-cloud infrastructure—supervisory control and data acquisition (SCADA) systems, industrial programmable logic controllers, and commercial smart meters—whose cost runs to many thousands of pounds. This places intelligent energy management firmly out of reach of small and medium-sized enterprises (SMEs), even though SMEs collectively account for a large share of industrial energy use. Moreover, much of the existing research addresses individual components in isolation—a forecasting model here [17], an anomaly detector there [18]—rather than demonstrating a complete, reproducible pipeline. As a result, an affordable and openly reproducible end-to-end system that an under-resourced organisation could actually adopt has remained largely unaddressed.

This paper addresses that gap. It presents the design, implementation, and evaluation of a low-cost smart grid sensor network for industrial energy management that integrates IoT sensing with machine learning in a single, reproducible

pipeline. In contrast to prior work that depends on costly proprietary hardware or that treats forecasting and anomaly detection separately, the proposed system combines inexpensive embedded sensing (targeting hardware of the order of £30), a documented and version-controlled data pipeline, an LSTM-based demand forecaster, an Isolation Forest anomaly detector, and a real-time monitoring dashboard. The work is grounded in a Design Science Research methodology [19], [20] and follows established guidance for the conduct of the research project [21].

The principal contributions of this paper are:

- an affordable, open-source, end-to-end architecture that brings together low-cost IoT sensing, wireless telemetry, time-series forecasting, unsupervised anomaly detection, and real-time visualisation;
- an empirical evaluation of an LSTM demand forecaster on a public benchmark dataset, including a comparison against naive and moving-average baselines; and
- a reproducible pipeline—with code, processed data, and trained models released publicly—that lowers the barrier to intelligent energy management for resource-constrained organisations.

The remainder of this paper is organised as follows. Section II defines the problem and its scale. Section III states the aim and objectives. Section IV reviews related work, critically compares existing approaches, and derives the set of parameters that informs the proposed solution architecture. Section V describes the methodology and system design, Section VI reports and discusses the results, and Section VII concludes and outlines future work.

II. PROBLEM STATEMENT

Industrial activity accounts for approximately one third of total global energy use [1]. A substantial part is consumed inefficiently, and in many facilities that inefficiency is never detected because consumption is not measured at the level of individual processes or machines.

The burden falls disproportionately on SMEs, which constitute the majority of industrial and commercial organisations yet mostly operate without intelligent monitoring. The principal barrier is cost: SCADA systems, industrial programmable logic controllers, and commercial smart-metering platforms typically require capital investment running to tens of thousands of pounds alongside specialist installation and maintenance [4], [14]. The consequence is that these businesses operate effectively “blind”: waste accumulates unseen, faults develop without early warning, and there is no basis on which to forecast demand.

Where monitoring exists it is frequently reactive: threshold alarms register a problem only after a value crosses a fixed limit, and are poorly suited to the gradual drift or context-dependent anomalies that precede failure [13].

III. AIM AND OBJECTIVES

Aim. To design, implement, and evaluate a low-cost smart grid sensor network for industrial energy management that

uses the Internet of Things and machine learning to deliver real-time monitoring, short-term demand forecasting, and anomaly detection, thereby demonstrating that capabilities previously confined to expensive proprietary systems can be achieved with affordable hardware and open-source software. The objectives that operationalise this aim are as follows:

- 1) To critically review the literature on IoT-based energy monitoring, machine-learning demand forecasting, and anomaly detection, in order to identify the research gap and define a precise project scope.
- 2) To prepare benchmark energy data—principally the UCI Individual Household Electric Power Consumption dataset [22]—through a documented, version-controlled pipeline applying chronological partitioning to prevent data leakage.
- 3) To design the IoT sensing architecture around a low-cost embedded platform using Bluetooth Low Energy and Wi-Fi telemetry [5], [23].
- 4) To develop and train a Long Short-Term Memory (LSTM) network [7] for short-term energy demand forecasting using TensorFlow/Keras [24], and to benchmark it against naive and moving-average baselines [25].
- 5) To implement unsupervised anomaly detection using an Isolation Forest model [8] that flags abnormal consumption patterns without requiring labelled fault data.
- 6) To build a real-time monitoring dashboard usable by a non-specialist operator, and to evaluate all models on held-out test data using MAE, RMSE, MAPE, and R^2 for forecasting and precision, recall, and F1 for detection, and to release the complete system as an openly reproducible artefact.

IV. LITERATURE REVIEW

This work draws on four overlapping bodies of research: IoT-based energy monitoring, machine-learning demand forecasting, anomaly detection, and low-cost embedded computing.

A. IoT-Based Energy Monitoring

Broad surveys of IoT-enabled smart grids [26] establish that sensing, communication, and analytics are now treated as a single integrated stack. At the application level, reviews of smart-home IoT [9] and smart-meter analytics [27] show that fine-grained measurement with feedback influences consumption, while Hafeez *et al.* [10] couple sensing with price-based demand response. At system scale, Silva *et al.* [4] address AC microgrids under grid and security constraints, Vijayalakshmi *et al.* [14] demonstrate energy management on embedded systems, and Bin Mofidul *et al.* [15] combine acquisition, anomaly detection, and monitoring across edge and cloud AI. Two limitations recur: attention concentrates on individual residences and utility-scale systems, leaving the industrial SME underserved; and the more capable integrated platforms achieve breadth through costly edge-cloud infrastructure, reintroducing the affordability barrier.

B. Machine Learning for Demand Forecasting

The field has shifted decisively from statistical models towards deep learning [6], with reviews tracing rapid adoption for load forecasting [12]. The LSTM [7] has become dominant for short-term load: Kong *et al.* [11] demonstrated its ability to capture household dynamics and Inala *et al.* [28] apply the same family in a smart-grid setting. Comparative studies refine this picture—Mahjoub *et al.* [29] benchmark LSTM against multi-layer and drop-GRU variants and expose an accuracy–efficiency trade-off, while Wen *et al.* [17] pursue accuracy through a hybrid model at the cost of complexity. Crucially, Makridakis *et al.* [25] found that machine-learning forecasters do not automatically outperform simpler methods and argued for evaluation against strong baselines—a finding treated here as binding rather than advisory. The broader gap is that most studies optimise accuracy on a single dataset and report little on deployability: model size, inference cost, or suitability for constrained hardware.

C. Anomaly Detection in Energy Systems

Himeur *et al.* [13] review AI-based anomaly detection in buildings and identify label scarcity as the field’s central tension, dividing the literature into two camps. Supervised approaches— XGBoost with LSTM [30] and network-level detection [31]— can be highly accurate but presuppose labelled examples of the very faults they aim to catch. Unsupervised methods avoid that dependency: the Isolation Forest [8] isolates anomalies efficiently, clustering has been applied to energy series [32], and variational recurrent autoencoders achieve strong results at considerable computational weight [18]. Decisively for this work, Jamshidi *et al.* [33] pair an autoencoder with an Isolation Forest specifically for resource-constrained IoT edge gateways, confirming the Isolation Forest belongs at the lightweight end of the spectrum.

D. Low-Cost Embedded and Edge Computing

TinyML [23] established that machine-learning inference can run on low-power microcontrollers, and applied studies confirm this: most directly relevant, Baskar *et al.* [34] deploy the Arduino Nano 33 BLE Sense, the platform adopted here, for industrial wastewater monitoring. The remaining gap is integrative: edge-ML studies typically address a single task and rarely combine forecasting, detection, and visualisation into one reproducible system.

E. Synthesis and Proposed Architecture

Taken together, the literature establishes four points. IoT monitoring is mature but concentrated away from the industrial SME. The LSTM is effective but must be validated against simple baselines and is rarely assessed for deployability. Lightweight unsupervised detection is the natural fit where labels are scarce and compute is limited. Low-cost edge hardware is proven for sensing. What remains absent is a single affordable, openly reproducible system integrating all four; Ahmed *et al.* [16] move toward such integration but not at the SME-accessible price point this project targets.

Smart Grid ML System Architecture

Based on typical ML project architecture adapted from Smart Grid energy management

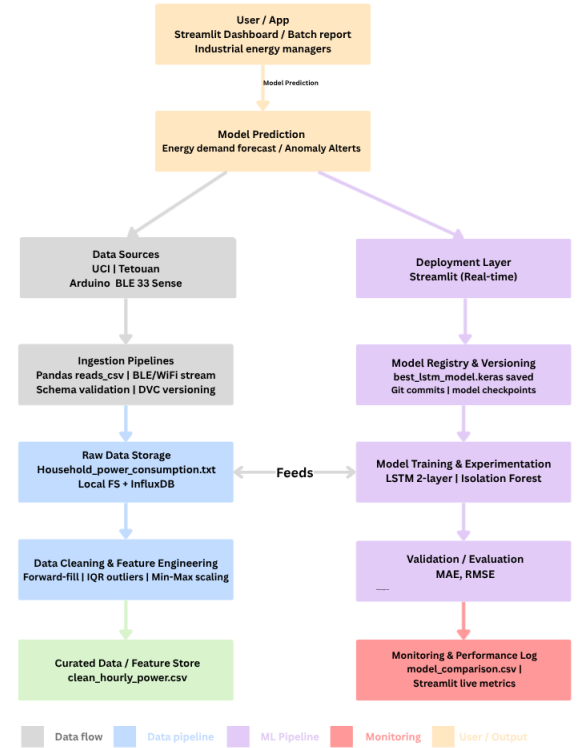


Fig. 1: Proposed four-layer system architecture.

These findings yield the design parameters that govern the system: affordability, realised as a ~£30 embedded platform; baseline validation, realised as mandatory naive and moving-average comparison; a label-free anomaly paradigm, realised as an Isolation Forest chosen over deep autoencoders for its computational footprint; leakage-free evaluation, realised as a chronological split with training-only scaler fitting; and reproducibility, realised as public release of code, data, and models. Guided by these, the proposed system is the four-layer architecture of Fig. 1: an IoT sensing layer on the validated low-cost platform [34]; a data-pipeline layer providing leakage-free preprocessing; a machine-learning layer pairing an LSTM forecaster [11] with an Isolation Forest detector [8]; and a visualisation layer delivering a real-time dashboard. Compared with the edge–cloud system of Bin Mofidul *et al.* [15], the design deliberately trades infrastructural sophistication for affordability and reproducibility.

V. METHODOLOGY AND SYSTEM DESIGN

A. Research Approach

The project follows a Design Science Research (DSR) methodology [19], [20], in which knowledge is generated through the construction and rigorous evaluation of an artefact—here, the integrated smart grid monitoring system. Because the requirements were exploratory and the work

was research-intensive, an incremental and iterative process was adopted, comprising four phases: data acquisition and preparation, model development, evaluation against baselines and integration of the models into a monitoring dashboard. The physical sensing layer follows the low-cost architecture set out in Section IV; the evaluation reported here uses two public benchmark datasets as its data source, with live hardware deployment reserved for future work.

B. Dataset and Preprocessing

The principal source of secondary data is the UCI Individual Household Electric Power Consumption dataset [22], comprising 2,075,259 minute-level measurements recorded between December 2006 and November 2010. It contains seven electrical quantities: global active power, global reactive power, voltage, global intensity, and three sub-metering channels. This dataset was selected because it provides a long, continuous, real-world record whose daily, weekly, and seasonal structure is well suited to sequence modelling. It is worth noting that other public benchmarks target related but distinct problems; the Smart Grid Stability dataset [35], for example, poses a binary stability-classification task rather than continuous demand forecasting, and is therefore outside the scope of the forecasting model developed here.

The raw data were prepared through a documented, version-controlled pipeline. First, 25,979 missing values were imputed by forward-fill, the standard choice for time series in which the most recent observation best estimates the next. Second, the series was resampled from one-minute to hourly resolution, producing 34,589 clean records and reducing noise while preserving the temporal patterns relevant to energy management. Third, extreme values were screened using the interquartile-range method, and all features were normalised to $[0, 1]$ using min-max scaling as required for stable recurrent-network training. Fourth, a sliding window of 24 hours was applied, yielding 34,565 supervised input-output sequences. Finally, the data were partitioned chronologically into training, validation, and test sets in a 70/15/15 ratio (24,195, 5,184, and 5,186 samples respectively), with the scaler fitted on the training partition only so that no information from future time steps could leak into the model input [25].

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \quad (1)$$

where $x_{\min}$ and $x_{\max}$ are computed on the training partition only. The forecasting task was then framed as a supervised sliding-window problem: given the previous $w = 24$ hourly observations, the model predicts the next hour,

$$\hat{y}_t = f(x_{t-w}, x_{t-w+1}, \dots, x_{t-1}), \quad (2)$$

where $f(\cdot)$ is the trained LSTM.

C. Demand Forecasting Model

Short-term demand forecasting was addressed with a Long Short-Term Memory (LSTM) network [7], whose gating

mechanism is well matched to the multi-scale temporal dependencies in energy data and which has an established record in load forecasting [11]. At each time step t , an LSTM unit updates its cell state C_t and hidden state h_t from the current input x_t and the previous hidden state h_{t-1} through three gates—forget (f_t), input (i_t), and output (o_t)—defined as

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \quad (3)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \quad (4)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C), \quad (5)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \quad (6)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \quad (7)$$

$$h_t = o_t \odot \tanh(C_t), \quad (8)$$

where $W_{\{f,i,C,o\}}$ and $b_{\{f,i,C,o\}}$ are the learned weight matrices and bias vectors, $\sigma(\cdot)$ is the logistic sigmoid, $\tanh(\cdot)$ is the hyperbolic tangent, and $\odot$ denotes element-wise multiplication. The forget gate (3) controls how much of the previous cell state is retained, the input gate (4) and candidate (5) determine what new information is written, and the output gate (7) produces the hidden state passed to the next step. This gating is what allows the network to retain information over the long daily and weekly cycles present in the energy series. The network comprises two stacked LSTM layers of 64 and 32 units, each followed by a dropout layer with rate 0.2, and a dense output neuron, giving 29,857 trainable parameters. It was trained to minimise the mean-squared-error loss using the Adam optimiser with a batch size of 32 for up to 50 epochs, with early stopping (patience of 10 epochs) to prevent overfitting; the implementation used TensorFlow/Keras [24] with a fixed random seed for reproducibility. To place the model's performance in context, two transparent baselines were implemented: a naive persistence predictor, which forecasts the next hour as equal to the current one, and a 24-hour moving average.

D. Anomaly Detection Model

Anomalous consumption was identified with an Isolation Forest [8], chosen because it is unsupervised—and therefore requires no labelled fault data, which is unavailable for this dataset—and because it is computationally lightweight and thus suited to low-cost deployment. The method builds an ensemble of trees that recursively partition the feature space along randomly selected features and split points, scoring each observation by the mean path length required to isolate it: anomalies, being few and readily separable, isolate in short paths, and a threshold set by the contamination parameter separates them from the remainder.

The detector operated on all seven electrical features simultaneously (multivariate approach), after standardising each feature to zero mean and unit variance so that no single quantity dominated by virtue of its scale. The model was configured with 200 base estimators, a contamination rate of 0.01, automatic subsampling, and a fixed random seed.

Because the dataset carries no ground-truth anomaly labels, a transparent statistical baseline—a Z-score rule flagging any hour whose global active power lies more than three standard deviations from the mean—was used as a reference point for comparison.

E. Evaluation Metrics

Forecasting accuracy was quantified on the held-out test set using the mean absolute error (MAE) and root-mean-square error (RMSE), both expressed in kilowatts for direct interpretability, together with the mean absolute percentage error (MAPE) and the coefficient of determination R^2 ; the LSTM was compared against the naive and moving-average baselines on identical data. For anomaly detection, in the absence of ground-truth labels, precision, recall, and the F1-score were computed relative to the Z-score reference, and therefore measure agreement between the two methods rather than correctness against true anomalies. They are complemented by an analysis of the multivariate anomalies the Isolation Forest detects but the univariate baseline misses. Reporting over a bounded window necessarily characterises typical rather than worst-case performance, a constraint revisited in Section VI.

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad (9)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}, \quad (10)$$

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right|, \quad (11)$$

$$R^2 = 1 - \frac{\sum_{t=1}^n (y_t - \hat{y}_t)^2}{\sum_{t=1}^n (y_t - \bar{y})^2}, \quad (12)$$

where y_t is the actual value, $\hat{y}_t$ the prediction, $\bar{y}$ the mean of the actual values, and n the number of test observations. For the anomaly detector, agreement with the Z-score reference was quantified using precision, recall, and their harmonic mean, the F1-score,

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}, \quad (13)$$

where TP , FP , and FN are the counts of records flagged by both methods, by the Isolation Forest only, and by the Z-score reference only, respectively. The Z-score used by the baseline is

$$z_t = \frac{x_t - \mu}{\sigma}, \quad (14)$$

with μ and σ the mean and standard deviation of global active power; an hour is flagged when $|z_t| > 3$.

TABLE I: Forecasting performance on the held-out test set (lower is better, except R^2)

Model	MAE (kW)	RMSE (kW)	MAPE (%)	R^2
Naive	0.4054	0.6350	44.07	0.2361
Moving Average	0.5768	0.7066	87.89	0.0542
LSTM	0.3707	0.5421	45.77	0.4432

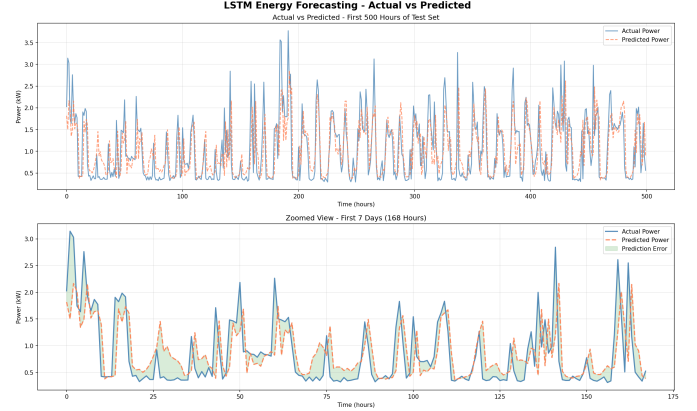


Fig. 2: LSTM forecast against actual demand on the UCI test set.

F. Implementation and Reproducibility

The system was implemented in Python 3.11 using TensorFlow/Keras, scikit-learn, pandas, and NumPy. Development and training were carried out on a consumer laptop (AMD Ryzen 7 6800U, 16 GB RAM), with the option of GPU-accelerated training on a cloud platform for larger workloads. The visualisation layer is a Streamlit dashboard that loads the saved model outputs and presents demand forecasts alongside anomaly alerts. To support full reproducibility, fixed random seeds were used throughout, and all source code, the processed dataset, trained model files, and results have been released in a public repository.

VI. RESULTS AND DISCUSSION

This section reports the performance of the two models on the held-out data and then discusses their practical significance and limitations. Unless otherwise stated, forecasting metrics are reported over a 500-hour window of the held-out test set, consistent with the figures presented in the monitoring dashboard.

A. Demand Forecasting Performance

Table I reports the LSTM alongside the two baselines. The LSTM achieved the lowest error on the metrics appropriate to this task, with an MAE of 0.3707 kW and an RMSE of 0.5421 kW—within roughly 370 W of true hourly demand—reducing MAE by 8.6% against naive persistence and 35.7% against the moving average, and explaining 44.3% of the variance ($R^2 = 0.4432$) against 23.6% and 5.4% respectively. Figure 2 shows the forecast tracking the actual series.

The LSTM's MAPE of 45.77% marginally exceeds naive's 44.07% despite clear superiority on every other measure. This

TABLE II: Agreement between the Isolation Forest and the Z-score baseline

Category	Records
Flagged by Isolation Forest	346
Flagged by Z-score baseline	440
Flagged by both	267
Isolation Forest only (multivariate)	79
Z-score only (univariate)	173

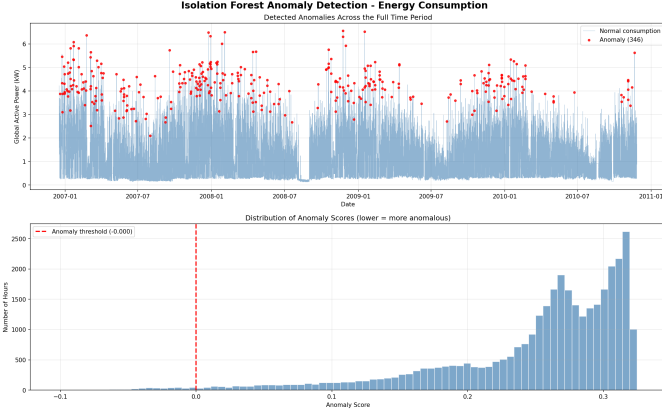


Fig. 3: Detected anomalies across the UCI series (top) and the anomaly-score distribution (bottom).

reflects a known weakness of MAPE on such data: household consumption falls close to zero overnight, and dividing by near-zero values inflates percentage error irrespective of absolute accuracy. MAPE is therefore not a reliable discriminator here. The moving average performs worst throughout, its smoothing lagging the peaks the LSTM anticipates. Placed against the wider literature, these figures are consistent with established findings rather than anomalous. Kong *et al.* [11] report comparable advantages for LSTM over statistical baselines on residential load, and the modest household margin observed here reflects the same ceiling they identify: single-dwelling demand is dominated by stochastic appliance events that no model can anticipate from load history alone. The far larger margin on aggregated data is likewise the behaviour the forecasting reviews [12] attribute to demand aggregation, and it is the reason Makridakis *et al.* [25] insist on baseline comparison rather than cross-study comparison of absolute error.

B. Anomaly Detection Performance

The Isolation Forest, operating on all seven electrical features, flagged 346 of the 34,589 hourly records (1.00%) as anomalous; Figure 3 shows their distribution across the series and in score space. Against the univariate Z-score baseline, which flagged 440 records, the methods agreed on 267, the Isolation Forest alone flagged 79, and the Z-score rule alone flagged 173 (Table II).

Treating the Z-score flags as a reference gives a precision of 0.7717, a recall of 0.6068, and an F1-score of 0.6794.

TABLE III: Forecasting performance on the Tetouan City test set

Model	MAE (kW)	RMSE (kW)	MAPE (%)	nMAE (%)
Naive	4,007.33	5,465.53	6.24	6.21
Moving Average	11,695.04	14,034.60	18.92	18.12
LSTM	1,521.60	2,299.07	2.41	2.36

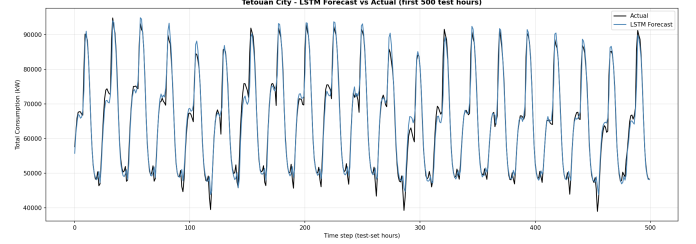


Fig. 4: LSTM forecast against actual city-scale demand on the Tetouan test set.

These figures measure *agreement* between two methods rather than correctness against true anomalies, since no ground-truth labels exist; the Z-score rule is a transparent reference, not an authority. Read this way, the Isolation Forest reproduces the anomalies a simple statistical rule catches, while its distinctive value lies in the 79 additional records—unusual *combinations* such as elevated intensity and sub-metering activity without an extreme total-power value, which a single-feature rule cannot detect by construction.

C. Cross-Dataset Validation on the Tetouan City Dataset

To assess generalisation beyond a single household, the complete workflow was re-applied, unchanged, to the Power Consumption of Tetouan City dataset: 52,416 ten-minute records for 2017 across three distribution zones of Tetouan, Morocco, with meteorological variables. The data were resampled to hourly resolution (8,736 records, no missing values), zone loads aggregated into total city consumption, and the same 24-hour window and chronological 70/15/15 split applied (6,115 training, 1,310 validation, 1,311 test samples) with identical model configuration.

Table III reports the results. The LSTM achieved an MAE of 1,521.60 kW against 4,007.33 kW for naive persistence and 11,695.04 kW for the moving average—improvements of 62.0% and 87.0%—with a MAPE of 2.41% and a normalised MAE of 2.36% of mean load; Figure 4 shows the close tracking of the smooth city-scale series. Accuracy is markedly higher than on household data, as expected: a city-scale load aggregates thousands of consumers into a smooth, strongly periodic series, whereas a single household is dominated by irregular appliance events. MAPE is also reliable here because city-scale consumption never approaches zero.

The detector was likewise re-applied, operating multivariately on total consumption with the meteorological features. It flagged 88 of the 8,736 records, of which only 7 coincided with the 35 flagged by the Z-score rule ($F1 = 0.1138$). This low agreement is expected by construction rather than indicative of failure: a smooth seasonal load produces few $|z| > 3$

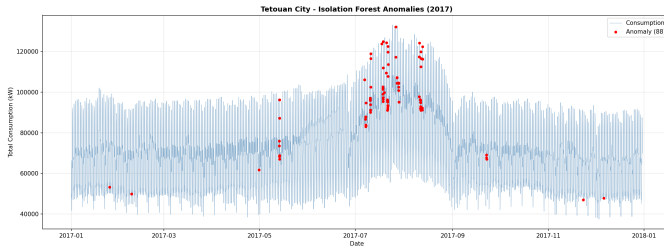


Fig. 5: Isolation Forest detections on the Tetouan series, concentrated in summer demand peaks.

extremes, while the multivariate detector responds to unusual *weather-demand combinations* invisible to a consumption-only threshold. Consistent with this, the 81 records flagged only by the Isolation Forest concentrate in July and August during afternoon and evening hours—heat-driven peaks in which high temperature and high load co-occur atypically, visible in Figure 5.

D. Discussion

The results support the central claim of this work: that capable energy analytics can be delivered with lightweight, low-cost methods rather than proprietary infrastructure. Three findings deserve emphasis.

First, the improvement over baselines is modest on household data and substantial on aggregated data, and this asymmetry is informative rather than incidental. A single household is dominated by stochastic appliance events that no model can anticipate from load history alone, so the achievable ceiling is low; a city-scale series aggregates thousands of such events into a smooth, strongly periodic signal in which the LSTM’s capacity to model multi-scale temporal structure is fully exercised. The practical implication for an SME is encouraging: an industrial facility with many machines running to a production schedule sits closer to the aggregated regime than to the single-household one, so the accuracy demonstrated on Tetouan is the more representative indication of what such a deployment would achieve. This behaviour is consistent with the established record of the LSTM on **short-term load forecasting** [11] and with the caution of Makridakis *et al.* [25] that superiority must be demonstrated rather than assumed—a caution treated here as binding, which is why the baselines were evaluated on identical data rather than quoted from the literature.

Second, the two detection regimes are complementary rather than competing. On volatile household data the Isolation Forest largely reproduces what a univariate statistical rule already catches, adding a margin of 79 combination anomalies; on smooth aggregated data almost its entire contribution is multivariate, since few observations are extreme in any single feature. A practitioner should therefore expect the value of multivariate detection to increase precisely where simple thresholds become least informative—the regime an industrial deployment is most likely to occupy. That the detector required

no labelled faults is what makes it deployable at all, given that such labels are rarely available in practice [13].

Third, affordability was achieved without a corresponding sacrifice in capability. The forecaster carries 29,857 parameters and trained in approximately eleven minutes on a consumer laptop CPU with no GPU at any stage; both models are small enough to be compatible with the low-cost deployment the project targets, and the complete pipeline runs in a working dashboard.

Several limitations qualify these conclusions. Forecasting metrics are reported over a bounded test window rather than the full multi-year series, so they characterise typical rather than worst-case performance. The anomaly evaluation, lacking labelled faults, is an agreement study rather than a measurement of detection accuracy, and its precision and recall should not be read as such. Neither dataset is industrial: the cross-dataset validation establishes transferability across load scales but not across load *types*, and validation on live industrial sensor data remains future work. Finally, both models are black boxes, and post-hoc explanation would be necessary before an operator could be expected to act on a flagged hour with confidence. These limitations are revisited in Section VII.

VII. CONCLUSION AND FUTURE WORK

This paper set out to demonstrate that intelligent energy management—real-time monitoring, short-term demand forecasting, and anomaly detection—can be delivered at a cost accessible to small and medium-sized enterprises, using low-cost hardware and open-source software rather than proprietary infrastructure. An end-to-end pipeline was designed, implemented, and evaluated: a documented, leakage-free preparation stage reducing 2,075,259 raw observations to 34,589 hourly records; a two-layer LSTM forecaster of 29,857 parameters benchmarked against transparent baselines; a multivariate Isolation Forest requiring no labelled faults; and a dashboard surfacing both. The complete system was released publicly for independent replication.

The empirical results support the central claim. On the UCI benchmark the LSTM achieved an MAE of 0.3707 kW—within roughly 370 W of true hourly demand—reducing error by 8.6% against naive persistence and 35.7% against a 24-hour moving average, while explaining 44.3% of the variance against 23.6% for naive. The Isolation Forest flagged 346 records, 267 agreeing with a Z-score reference and 79 identified by the multivariate detector alone. Cross-dataset validation on the Tetouan City load confirmed that the unchanged pipeline transfers across scales, with improvements of 62.0% and 87.0% and a normalised MAE of 2.36%. The entire evaluation ran on a consumer laptop CPU, with training completing in approximately eleven minutes—its part of the affordability argument.

The work has limitations that define its future direction. The evaluation is grounded in secondary data from two public benchmarks, as defined by the scope of this project; extending it to primary data from the designed sensing layer on the Arduino Nano 33 BLE Sense platform is a natural direction

for further work. The anomaly evaluation remains an agreement study absent labelled faults, and deployment feedback would allow the contamination parameter to be tuned against ground truth. Both models are presently black boxes; SHAP-based attribution [36] would let operators see *why* an hour was flagged. Finally, the forecaster could be extended from univariate to multivariate input, exploiting the meteorological variables the Tetouan analysis shows interact strongly with demand.

In sum, the contribution of this work is not a new algorithm but a demonstration: that established, lightweight methods, engineered carefully into a single reproducible system, are sufficient to bring smart-grid capabilities within reach of the organisations that currently lack them.

40

REFERENCES

- [1] I. E. Agency, "International energy agency (iea) world energy outlook 2022," *International Information Administration*, vol. 524, 2022.
- [2] F. Fan, P. Kulkarni, S. Gormus, C. Efthymiou, G. Kalogridis, M. Sooriyabandara, Z. Zhu, S. Lambotharan, and W. H. Chin, "Smart grid communications: Overview of research challenges, solutions, and standardization activities," *IEEE Communications Surveys & Tutorials*, vol. 15, no. 1, pp. 21–38, 2012.
- [3] Y. Yan, Y. Qian, H. Sharif, and D. Tipper, "A survey on smart grid communication infrastructures: Motivations, requirements and challenges," *IEEE communications surveys & tutorials*, vol. 15, no. 1, pp. 5–20, 2012.
- [4] A. A. Silva, J. C. López, C. P. Guzman, N. B. Arias, M. J. Rider, and L. C. da Silva, "An iot-based energy management system for ac microgrids with grid and security constraints," *Applied energy*, vol. 337, p. 120904, 2023.
- [5] S. Munirathinam, "Industry 4.0: Industrial internet of things (iiot)," in *Advances in computers*. Elsevier, 2020, vol. 117, no. 1, pp. 129–164.
- [6] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [7] S. Hochreiter, "Long short-term memory," *Neural Computation MIT-Press*, 1997.
- [8] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *2008 eighth IEEE international conference on data mining*. IEEE, 2008, pp. 413–422.
- [9] M. Alaa, A. A. Zaidan, B. B. Zaidan, M. Talal, and M. L. M. Kiah, "A review of smart home applications based on internet of things," *Journal of network and computer applications*, vol. 97, pp. 48–65, 2017.
- [10] J. Hafeez, Z. Wadud, I. U. Khan, I. Khan, Z. Shafiq, M. Usman, and M. U. A. Khan, "Efficient energy management of iot-enabled smart homes under price-based demand response program in smart grid," *Sensors*, vol. 20, no. 11, p. 3155, 2020.
- [11] W. Kong, Z. Y. Dong, Y. Jia, D. J. Hill, Y. Xu, and Y. Zhang, "Short-term residential load forecasting based on lstm recurrent neural network," *IEEE transactions on smart grid*, vol. 10, no. 1, pp. 841–851, 2017.
- [12] M. Q. Raza and A. Khosravi, "A review on artificial intelligence based load demand forecasting techniques for smart grid and buildings," *Renewable and Sustainable Energy Reviews*, vol. 50, pp. 1352–1372, 2015.
- [13] J. Himeur, K. Ghanem, A. Alsalemi, F. Bensaali, and A. Amira, "Artificial intelligence based anomaly detection of energy consumption in buildings: A review, current trends and new perspectives," *Applied Energy*, vol. 287, p. 116601, 2021.
- [14] V. Vijayalakshmi, R. Raman, G. Venkatesh, C. J. Rawandale, G. Kalaimani, and C. Srinivasan, "Smart energy management using iot-based embedded systems," in *2023 International Conference on Sustainable Communication Networks and Application (ICSCNA)*. IEEE, 2023, pp. 299–304.
- [15] R. Bin Mofidul, M. M. Alam, M. H. Rahman, and Y. M. Jang, "Real-time energy data acquisition, anomaly detection, and monitoring system: Implementation of a secured, robust, and integrated global iiot infrastructure with edge and cloud ai," *Sensors*, vol. 22, no. 22, p. 8980, 2022.
- [16] A. Ahmed, H. Ansari, B. Khan, Z. Ullah, S. M. Ali, C. A. A. Mehmood, M. B. Qureshi, I. Hussain, M. Jawad, M. U. S. Khan *et al.*, "Machine learning based energy management model for smart grid and renewable energy districts," *IEEE Access*, vol. 8, pp. 185 059–185 078, 2020.
- [17] X. Wen, J. Liao, Q. Niu, N. Shen, and Y. Bao, "Deep learning-driven hybrid model for short-term load forecasting and smart grid information management," *Scientific reports*, vol. 14, no. 1, p. 13720, 2024.
- [18] J. Pereira and M. Silveira, "Unsupervised anomaly detection in energy time series data using variational recurrent autoencoders with attention," in *2018 17th IEEE international conference on machine learning and applications (ICMLA)*. IEEE, 2018, pp. 1275–1282.
- [19] R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research1," *MIS quarterly*, vol. 28, no. 1, pp. 75–106, 2004.
- [20] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, "A design science research methodology for information systems research," *Journal of management information systems*, vol. 24, no. 3, pp. 45–77, 2007.
- [21] O. Zina *et al.*, *The essential guide to doing your research project*. Sage, 2021.
- [22] J. Hebrail and A. Berard, "Individual household electric power consumption data set. uci machine learning repository," *University of California, School of Information and Computer*, 2012.
- [23] J. Warden and D. Situnayake, *Tinyml: Machine learning with tensorflow lite on arduino and ultra-low-power microcontrollers*. O'Reilly Media, 2019.
- [24] M. Abadi, P. Barham, J. Chen, Z. Chen, A. Davis, J. Dean, M. Devin, S. Ghemawat, G. Irving, M. Isard *et al.*, "{TensorFlow}: a system for {Large-Scale} machine learning," in *12th USENIX symposium on operating systems design and implementation (OSDI 16)*, 2016, pp. 265–275.
- [25] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "Statistical and machine learning forecasting methods: Concerns and ways forward," *InfoSys one*, vol. 13, no. 3, p. e0194889, 2018.
- [26] A. Goudarzi, F. Ghayoor, M. Waseem, S. Fahad, and I. Traore, "A survey on iot-enabled smart grids: emerging, applications, challenges, and outlook," *Energies*, vol. 15, no. 19, p. 6984, 2022.
- [27] A. Yildiz, J. I. Bilbao, J. Dore, and A. B. Sproul, "Recent advances in the analysis of residential electricity consumption and applications of smart meter data," *Applied Energy*, vol. 208, pp. 402–427, 2017.
- [28] P. Inala, S. Gaddam, S. Etti, J. Karangula, and P. Kashetty, "Enhancing the power load prediction using lstm in a smart grid scenario for household data," in *2025 IEEE 1st International Conference on Smart and Sustainable Developments in Electrical Engineering (SSDEE)*. IEEE, 2025, pp. 01–06.
- [29] S. Hajjoub, L. Chrifi-Alaoui, B. Marhic, and L. Delahoche, "Predicting energy consumption using lstm, multi-layer gru and drop-gru neural networks," *Sensors*, vol. 22, no. 11, p. 4062, 2022.
- [30] Z. Chen, Z. Li, J. Huang, S. Liu, and H. Long, "An effective method for anomaly detection in industrial internet of things using xgboost and lstm," *Scientific reports*, vol. 14, no. 1, p. 23969, 2024.
- [31] K. Sahu and I. Mukherjee, "Machine learning based anomaly detection for iot network:(anomaly detection in iot network)," in *2020 4th international conference on trends in electronics and informatics (ICOEI)(48184)*. IEEE, 2020, pp. 787–794.
- [32] A. Jesmeen, J. Hossen, and A. B. A. Aziz, "Unsupervised anomaly detection for energy consumption in time series using clustering approach," *Engineering Science Journal*, vol. 5, no. 6, pp. 840–854, 2021.
- [33] A. Hamshidi, F. Erfan, O. Abdul-Wahab, M. Bellaiche, and F. Khomh, "Lightweight autoencoder-isolation forest anomaly detection for green iot edge gateways," *arXiv preprint arXiv:2511.18235*, 2025.
- [34] S. Baskar, V. Anuprapaa *et al.*, "Edge-powered monitoring with arduino nano ble sense: Real-time industrial wastewater quality assessment," in *2024 International Conference on Communication, Computing, Smart Materials and Devices (ICCCSMD)*. IEEE, 2024, pp. 1–4.
- [35] A. Arzamasov, K. Böhm, and P. Jochem, "Towards concise models of grid stability," in *2018 IEEE international conference on communications, control, and computing technologies for smart grids (SmartGrid-Com)*. IEEE, 2018, pp. 1–6.
- [36] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in neural information processing systems*, vol. 30, 2017.

● 24% Overall Similarity

Top sources found in the following databases:

- 24% Internet database
- 0% Publications database
- 0% Submitted Works database

TOP SOURCES

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	arxiv.org Internet	3%
2	oulurepo.oulu.fi Internet	1%
3	assets-eu.researchsquare.com Internet	<1%
4	sci-cult.net Internet	<1%
5	koreascience.kr Internet	<1%
6	ideals.illinois.edu Internet	<1%
7	ajss.univ-boumerdes.dz Internet	<1%
8	bvrit.ac.in Internet	<1%
9	jetty.klnpa.org Internet	<1%

10	strathprints.strath.ac.uk Internet	<1%
11	techscience.com Internet	<1%
12	researchgate.net Internet	<1%
13	d197for5662m48.cloudfront.net Internet	<1%
14	export.arxiv.org Internet	<1%
15	mdpi-res.com Internet	<1%
16	ijsr.net Internet	<1%
17	sigma-not.pl Internet	<1%
18	journal.uty.ac.id Internet	<1%
19	downloads.hindawi.com Internet	<1%
20	theses.whiterose.ac.uk Internet	<1%
21	iris.polito.it Internet	<1%

22	eprints.utm.edu.my Internet	<1%
23	pubs2.ascee.org Internet	<1%
24	thesesjournal.com Internet	<1%
25	res.cloudinary.com Internet	<1%
26	ntnuopen.ntnu.no Internet	<1%
27	eprints.gla.ac.uk Internet	<1%
28	s3-ap-southeast-1.amazonaws.com Internet	<1%
29	frontiersin.org Internet	<1%
30	ijeecs.iaescore.com Internet	<1%
31	opus.lib.uts.edu.au Internet	<1%
32	deepai.org Internet	<1%
33	file.techscience.com Internet	<1%

34	acr-journal.com Internet	<1%
35	d1q2ihs98wnwia.cloudfront.net Internet	<1%
36	uetpeshawar.edu.pk Internet	<1%
37	d-nb.info Internet	<1%
38	unige.iris.cineca.it Internet	<1%
39	politesi.polimi.it Internet	<1%
40	cris.unibo.it Internet	<1%
41	wseas.com Internet	<1%
42	informatica.si Internet	<1%
43	fnasjournals.com Internet	<1%
44	hal.science Internet	<1%
45	backend.orbit.dtu.dk Internet	<1%

46	journals.mesopotamian.press	Internet	<1%
47	pdfcoffee.com	Internet	<1%
48	opus.fhv.at	Internet	<1%
49	research-information.bris.ac.uk	Internet	<1%
50	vigir.missouri.edu	Internet	<1%
51	mdpi.com	Internet	<1%
52	chatmed-project.eu	Internet	<1%
53	tudr.thapar.edu	Internet	<1%
54	diva-portal.org	Internet	<1%
55	eprints.lancs.ac.uk	Internet	<1%